

LONGZHOU, China

WMO: 594170

Lat: 22.3398N

Lon: 106.8536E

Elev: 129

StdP: 99.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.0	(c) 7.6	(d) -0.5	(e) 3.7	(f) 13.9	(g) 1.6	(h) 4.3	(i) 12.8	(j) 4.9	(k) 17.9	(l) 4.3	(m) 16.7	(n) 1.0	(o) 110	(p) 0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.3	(c) 36.0	(d) 26.4	(e) 35.0	(f) 26.3	(g) 34.1	(h) 26.1	(i) 27.9	(j) 33.1	(k) 27.5	(l) 32.4	(m) 27.1	(n) 31.8	(o) 2.0	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.6	29.9	26.3	22.1	29.6	26.0	21.7	29.4	90.5	33.3	88.6	32.6	87.0	32.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.5	3.2	3.5	38.3	1.5	1.1	2.4	39.1	1.6	39.7	0.7	40.3	-0.4	41.0		
			3.1	28.7	1.4	0.6	2.1	29.1	1.3	29.5	0.5	29.8	-0.5	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	14.7	16.8	19.7	24.5	27.5	28.9	29.0	28.7	27.5	24.8	20.7	16.1
	DBStd	6.03	4.01	4.80	4.16	3.70	2.68	1.80	1.70	1.64	1.99	2.53	3.06	3.45
	HDD10.0	7	4	2	0	0	0	0	0	0	0	0	0	1
	HDD18.3	346	128	81	33	3	0	0	0	0	0	1	12	88
	CDD10.0	4853	151	193	302	434	542	566	590	581	526	459	319	191
	CDD18.3	2150	16	39	77	187	283	316	332	322	276	202	82	19
	CDH23.3	20666	58	219	409	1501	2926	3607	3769	3451	2676	1508	454	88
(13)	CDH26.7	8184	7	59	117	518	1234	1543	1610	1424	1050	507	104	12
(14)	Wind	WSAvg	1.2	1.4	1.5	1.4	1.3	1.2	1.2	1.1	1.1	1.1	1.1	1.2
(15) Precipitation	PrecAvg	1314	28	31	49	99	179	220	210	222	127	89	41	19
	PrecMax	1755	88	122	172	212	405	570	448	442	254	212	139	91
	PrecMin	950	0	0	8	4	57	41	66	47	64	2	1	0
	PrecStd	193	24	27	40	50	81	107	87	105	48	51	37	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	31.4	32.9	36.1	37.5	36.9	36.8	36.2	35.4	33.4	31.1	28.2
		MCWB	20.1	20.9	22.9	24.5	26.3	27.1	27.1	26.5	25.7	23.9	22.8	21.5
	2%	DB	24.9	28.5	29.7	33.2	35.4	35.4	35.4	35.1	34.1	32.0	29.3	25.8
		MCWB	19.2	21.0	22.8	24.9	26.1	26.9	26.8	26.5	25.4	23.7	22.1	19.9
	5%	DB	22.8	26.0	27.3	31.4	33.8	34.1	34.3	34.0	33.0	30.9	27.8	23.7
		MCWB	18.1	20.2	22.1	24.5	25.6	26.7	26.6	26.5	25.3	23.1	20.8	18.1
	10%	DB	20.8	23.8	25.3	29.6	32.2	32.9	33.1	32.8	31.8	29.7	26.0	21.8
		MCWB	17.2	19.4	21.2	23.8	25.2	26.5	26.6	26.4	24.9	22.7	20.3	17.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	22.5	24.2	26.4	27.9	28.3	28.4	28.2	27.5	25.9	24.5	22.6
		MCDB	25.1	28.2	29.6	32.1	33.7	33.7	33.9	33.3	32.0	30.9	28.3	26.8
	2%	WB	19.8	21.6	23.3	25.6	27.1	27.7	27.8	27.7	26.8	25.2	23.6	21.0
		MCDB	23.9	26.4	27.8	31.0	32.6	32.8	32.9	32.6	31.2	29.4	27.1	24.3
	5%	WB	18.9	20.7	22.6	25.0	26.6	27.3	27.3	27.2	26.4	24.6	22.7	19.6
		MCDB	22.1	25.0	26.5	29.8	31.6	32.0	32.1	31.8	30.5	28.3	25.9	22.5
	10%	WB	17.9	19.9	21.8	24.4	26.0	27.0	26.9	26.8	25.9	24.0	21.8	18.0
		MCDB	20.4	23.5	25.1	28.7	30.5	31.3	31.3	31.0	29.8	27.4	24.8	20.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.6	6.0	6.9	7.6	7.3	7.3	7.6	7.9	8.2	8.4	8.0
		MCDBR	10.6	10.7	9.7	9.6	10.2	9.4	9.5	9.6	9.6	10.0	10.9	10.6
		MCWBR	4.8	4.0	3.5	3.0	2.6	2.5	2.5	2.6	2.4	2.8	4.1	4.8
	5% WB	MCDBR	8.3	9.6	8.1	8.5	8.5	8.0	8.2	8.5	8.0	8.1	8.1	8.8
		MCWBR	4.1	4.0	3.5	3.0	2.9	2.7	2.7	2.9	2.5	2.8	3.6	4.6
(40) Clear-Sky Solar Irradiance	taub		0.597	0.621	0.792	0.746	0.561	0.542	0.513	0.577	0.617	0.711	0.595	0.620
	taud		1.718	1.706	1.432	1.556	2.008	2.076	2.150	1.948	1.818	1.586	1.776	1.689
	Ebn at Noon		663	684	593	636	761	769	792	745	706	608	661	622
	Edh at Noon		215	231	316	283	179	166	154	189	212	257	201	215
(44) All-Sky Solar Radiation	RadAvg		2.07	2.52	2.77	3.60	4.33	4.49	4.71	4.66	4.42	3.67	3.09	2.45
	RadStd		0.52	0.48	0.37	0.42	0.46	0.50	0.36	0.30	0.30	0.29	0.46	0.43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	N/A	+73

Nomenclature: See separate page